Coordinate Collapse in Utility-Weighted PCA Embeddings for Biometric Matching

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***Abstract*—Principal Component Analysis (PCA) is widely deployed for template generation and feature compression in biometric verification systems. While PCA orders its basis vectors by explained variance, matching metrics such as cosine similarity treat every coordinate of the resulting projection identically, assigning equal geometric importance to all retained dimensions. This isotropic treatment ignores the heterogeneous discriminative value of individual principal components. To address this discrepancy, we investigate XPCA, a utility-guided calibration framework that weights each principal component through an additive utility score integrating three complementary signals: normalized eigenvalue variance (V_i), class-discriminability via a regularized Fisher ratio (D_i), and statistical stability via a bootstrapped coefficient of variation (N_i). The utility score $S_i = \alpha V_i + \beta D_i - \gamma N_i$ is used to construct a diagonal calibration matrix $A = \mathrm{diag}(S_1, \ldots, S_k)$ applied to the PCA projections prior to cosine matching. Evaluating the framework on the IIT Delhi (IITD) Touchless Palmprint Database across subspace dimensions $k \in \{32, 64, 128, 256, 512\}$, we present the counter-intuitive finding that utility-based diagonal calibration *consistently degrades* matching performance compared to the standard PCA baseline. At $k = 128$, XPCA increases the Equal Error Rate (EER) from 9.50% to 13.17%. Through a systematic ablation study, hyperparameter sweep, and latency benchmarking, we demonstrate that this degradation is robust to all weight configurations, indicating a *structural* failure rather than suboptimal tuning. We derive the mathematical mechanism underlying this failure—*coordinate collapse*—proving that diagonal scaling under cosine similarity introduces quadratic amplification of the scaling coefficients. When utility scores are non-uniform, this quadratic distortion causes the metric tensor $M = A^2$ to become severely ill-conditioned ($\kappa(M) \to \infty$), effectively collapsing the embedding to a low-dimensional subspace dominated by a few high-utility coordinates. This destroys the distributed biometric information encoded across the full principal subspace. We formalize this result via the condition number of the induced Mahalanobis metric and validate the theory through ablation experiments. Finally, we outline regularized calibration alternatives—including softmax scaling, residual calibration, and rank-based weighting—that provably bound the condition number and may enable principled component weighting in future biometric systems.**

***Index Terms*—Biometric verification, principal component analysis, diagonal calibration, coordinate collapse, cosine similarity, metric learning, palmprint recognition, negative result.**

I. Introduction

Palmprint biometrics exploits the uniqueness of epidermal crease patterns, principal lines, and skin texture on the human palm for identity verification [2], [3]. The modality offers advantages over fingerprint recognition in terms of larger surface area and richer feature content, and over face recognition in terms of resistance to expression and aging variations. Modern deployments increasingly favor contactless, touchless acquisition to address hygiene and user-acceptance concerns [1]. However, touchless capture introduces systematic degradations—translation offsets, rotation variations, hand deformation, non-uniform illumination, and depth-of-field effects—that challenge matching accuracy [12].

To achieve real-time matching speeds and compact biometric templates on resource-constrained embedded platforms, the high-dimensional raw pixel representation ($D = 16{,}384$ for 128×128 images) must be projected into a compact, orthogonal feature space via dimensionality reduction. Principal Component Analysis (PCA) remains the dominant linear projection method in production biometric systems due to its computational efficiency, closed-form solution, and well-understood statistical properties [4], [6].

Standard PCA decomposes the dataset covariance matrix via eigendecomposition, ranking the orthogonal basis vectors (eigenvectors, or "EigenPalms" in the palmprint context) by their explained variance (eigenvalues). A core assumption is that higher variance implies higher biometric utility. However, this assumption is violated in practice: high-variance components often encode global illumination gradients and low-frequency spatial trends, while lower-variance components encode fine-grained crease patterns and ridge structures that carry richer class-discriminative information [7].

A second, independent limitation arises at the *matching* stage. After PCA projection, biometric templates are typically compared using cosine similarity or normalized correlation, metrics that treat the projected coordinate space as *isotropic*—each coordinate contributes equally to the angle computation. This isotropic treatment ignores the heterogeneous discriminative value of individual principal components.

These two observations—the misalignment between variance and discriminability, and the isotropic treatment of coordinates—motivate the intuitive idea of *component-wise calibration*: weighting each coordinate by a measure of its biometric utility before matching. Under this intuition, we designed and evaluated a framework called XPCA (eXplainable PCA), which computes an additive utility score S_i for each component combining three complementary signals—explained variance, Fisher-ratio class separability, and bootstrap stability—and applies the resulting weights as a diagonal scaling matrix.

The central finding of this paper is *negative*: utility-guided diagonal calibration *consistently degrades* verification performance across all tested configurations. At the optimal baseline dimension $k = 128$, the Equal Error Rate increases from 9.50% (unweighted PCA) to 13.17% (XPCA). This degradation is robust to all tested hyperparameter configurations, ablation settings, and dimensionalities.

Rather than treating this as a failed engineering exercise, we analyze the failure to uncover a fundamental structural limitation of diagonal calibration under angular matching metrics. Our theoretical contribution is the discovery and formal proof of the *coordinate collapse* mechanism: non-uniform diagonal scaling introduces quadratic amplification of the weight coefficients in the cosine similarity formulation, causing the induced metric tensor to become ill-conditioned and effectively collapsing the representation to a low-dimensional manifold.

Our specific contributions are:

1) A systematic evaluation of standard PCA versus utility-guided diagonal calibration (XPCA) across five subspace dimensions on the IITD touchless palmprint database, demonstrating consistent performance degradation.
2) A four-configuration ablation study and three-configuration hyperparameter sweep proving that the degradation is structural rather than a consequence of suboptimal tuning.
3) A formal mathematical derivation and proof of the coordinate collapse mechanism, showing that diagonal scaling under cosine similarity induces a Mahalanobis metric with unbounded condition number.
4) Concrete recommendations for regularized calibration formulations that provably bound the condition number of the metric tensor.

The remainder of this paper is organized as follows. Section II reviews related work on subspace methods, component weighting, and metric learning in biometrics. Section III presents the XPCA formulation, algorithmic pipeline, and computational complexity. Section IV describes the experimental protocol. Section V reports the experimental results. Section VI derives the coordinate collapse mechanism. Section VII discusses the scientific implications. Section VIII outlines regularized alternatives, and Section IX concludes.

II. Related Work

A. Subspace Methods in Biometrics

Turk and Pentland [5] established the "Eigenfaces" paradigm for face recognition, where PCA projections serve as holistic appearance features. Lu et al. [4] adapted this framework to palmprint recognition as "EigenPalms," demonstrating competitive verification accuracy with a simple nearest-neighbor matching protocol. Kumar [1] extended the approach to touchless palmprint acquisition, showing that PCA-based templates achieve robust verification under controlled conditions. These baseline studies treated the number of retained components k as a hyperparameter to be tuned via cross-validation, without analyzing how the matching metric interacts with the internal structure of the projected coordinates.

B. Discriminant Subspace Methods

Fisher Discriminant Analysis (FDA), also known as Linear Discriminant Analysis (LDA), addresses the variance-discriminability gap by projecting features onto axes that maximize the ratio of between-class to within-class scatter [7]. PCA+LDA cascades first reduce dimensionality with PCA to regularize the within-class scatter matrix, then apply LDA on the reduced coordinates. However, in biometric datasets with few training samples per class (the "small sample size" problem), the within-class scatter matrix is rank-deficient and its inverse is unstable [11]. In the IITD dataset used in this study, each class contains only 3 training images, making stable within-class scatter estimation extremely challenging and rendering PCA+LDA highly susceptible to overfitting.

C. Feature Weighting and Metric Learning

Feature weighting schemes attempt to assign importance coefficients to individual coordinates. Weighted PCA (WPCA) variants scale coordinates by inverse variance (whitening) to equalize coordinate contributions [6]. However, whitening amplifies high-frequency noise in low-variance directions, typically degrading verification performance.

Mahalanobis distance metric learning approaches—including Large Margin Nearest Neighbor (LMNN) [8], Information-Theoretic Metric Learning (ITML) [9], and the broader family of positive semi-definite metric learning methods—learn a matrix $M \succeq 0$ such that the Mahalanobis distance $d_M(x, y) = \sqrt{(x - y)^\top M(x - y)}$ respects class boundaries. When M is restricted to be diagonal, metric learning reduces to coordinate scaling, which is mathematically equivalent to the calibration scheme investigated in this paper. Our work reveals why such diagonal restrictions fail under angular (cosine) matching metrics, complementing the metric learning literature which predominantly uses Euclidean or Mahalanobis *distance* rather than *similarity* formulations.

D. Negative Results in Machine Learning

The machine learning community has increasingly recognized the scientific value of well-documented negative results [10]. Negative findings prevent redundant work, reveal implicit assumptions, and guide future research directions. In biometrics specifically, documenting why intuitively reasonable approaches fail provides essential design guidelines for practitioners. Our coordinate collapse finding falls into this category: it demonstrates that a mathematically natural idea (utility-based diagonal scaling) violates a structural requirement (bounded metric tensor conditioning) of the cosine similarity metric.

III. Proposed Method: XPCA Framework

A. Standard PCA Formulation

Let $X = [x_1, x_2, \ldots, x_N]^\top \in \mathbb{R}^{N \times D}$ be the training feature matrix, where N is the number of training samples and D is the raw pixel dimension. Let $\mu = \frac{1}{N}\sum_{n=1}^{N} x_n$ be the global training mean. The mean-centered training matrix is $\bar{X} = X - \mathbf{1}\mu^\top \in \mathbb{R}^{N \times D}$. The sample covariance matrix is:

$$\Sigma = \frac{1}{N}\bar{X}^\top \bar{X} \in \mathbb{R}^{D \times D} \tag{1}$$

In practice, SVD is applied to the mean-centered matrix:

$$\bar{X} = USV^\top \tag{2}$$

where the right singular vectors (columns of $V \in \mathbb{R}^{D \times D}$) are the eigenvectors of Σ, and the eigenvalues are $\lambda_i = s_i^2/N$. Retaining only the top-k components yields the projection matrix $W_k = [v_1, \ldots, v_k] \in \mathbb{R}^{D \times k}$. The projection of a centered test sample $x_c = x - \mu$ into the subspace is:

$$z = W_k^\top x_c \in \mathbb{R}^k \tag{3}$$

On the IITD dataset, the eigenvalue spectrum exhibits a steep exponential decay: the first component captures 11.78% of total variance ($\lambda_1 = 38.85$), while the 128th component captures only 0.096% ($\lambda_{128} = 0.317$). Cumulatively, 128 components retain 80.81% of the total variance, and 512 components retain 94.35%.

B. XPCA Utility Score Formulation

The XPCA framework computes an additive utility score S_i for each retained principal component $i \in \{1, \ldots, k\}$:

$$S_i = \alpha V_i + \beta D_i - \gamma N_i \tag{4}$$

where $\alpha, \beta, \gamma \geq 0$ are the weight hyperparameters controlling the contribution of each signal. The three component scores are defined as follows.

1) Normalized Variance Score (V_i): The proportion of the total energy (global variance) captured by component i:

$$V_i = \frac{\lambda_i}{\sum_{j=1}^{D} \lambda_j} \tag{5}$$

This score captures the standard PCA ranking criterion. By construction, $V_1 > V_2 > \ldots > V_k > 0$ and $\sum_{i=1}^{D} V_i = 1$.

2) Fisher Discriminability Score (D_i): This score measures the class-separability of component i using a stabilized Fisher ratio. Let $z_i^{(n)}$ denote the i-th coordinate of the PCA projection of training sample n, and let the class labels $c \in \{1, \ldots, C\}$ partition the training set. For each class c, compute the class mean $\mu_{c,i}$ and class variance $\sigma_{c,i}^2$ of the i-th coordinate. The Fisher score is:

$$D_i = \frac{S_B^{(i)}}{S_W^{(i)} + \epsilon} \tag{6}$$

where the between-class scatter on coordinate i is:

$$S_B^{(i)} = \frac{1}{C}\sum_{c=1}^{C} (\mu_{c,i} - \bar{\mu}_i)^2 \tag{7}$$

and the within-class scatter uses shrinkage regularization toward the global eigenvalue to prevent small-sample instability:

$$S_W^{(i)} = (1 - \lambda_{\text{shrink}}) \left(\frac{1}{C}\sum_{c=1}^{C} \sigma_{c,i}^2 \right) + \lambda_{\text{shrink}} \cdot \lambda_i \tag{8}$$

with shrinkage parameter $\lambda_{\text{shrink}} = 0.1$ and numerical regularization $\epsilon = 10^{-10}$. The shrinkage in Eq. 8 is essential because with only 3 training samples per class ($n_c = 3$), the raw within-class variance $\sigma_{c,i}^2$ is estimated from $n_c - 1 = 2$ degrees of freedom, making it highly volatile. The regularized estimate blends the raw within-class scatter with the global eigenvalue λ_i as a prior, stabilizing the Fisher ratio.

3) Bootstrap Instability Score (N_i): To quantify the *statistical stability* of the Fisher score, we use bootstrap resampling. We perform $T = 100$ bootstrap iterations. In each iteration t, the training set is resampled *with replacement*, and the Fisher score $D_i^{(t)}$ is recomputed on the resampled data. The instability of component i is measured by the coefficient of variation:

$$N_i = \frac{\text{std}_t \left(D_i^{(t)} \right)}{\text{mean}_t \left(D_i^{(t)} \right) + \epsilon} \tag{9}$$

Components with high N_i have Fisher scores that are sensitive to the training sample composition, indicating that their discriminability is not reliably estimated from the available data. By penalizing high-instability components (the $-\gamma N_i$ term in Eq. 4), the utility score favors components whose discriminability is both high and statistically robust.

C. Min-Max Normalization

To align the scales of the three heterogeneous scores, the raw utility vector $\mathbf{S} = [S_1, \ldots, S_k]$ is min-max normalized to $[0, 1]$:

$$S_i^{\text{norm}} = \frac{S_i - \min_j S_j}{\max_j S_j - \min_j S_j + \epsilon} \tag{10}$$

This normalization ensures that $S_{\min}^{\text{norm}} = 0$ and $S_{\max}^{\text{norm}} = 1$. As we prove in Section VI, this normalization is a direct contributor to the coordinate collapse failure because it forces at least one coordinate's weight to approach zero.

D. Diagonal Calibration and Matching

The normalized utility scores define a diagonal calibration matrix:

$$A = \text{diag}(S_1^{\text{norm}}, S_2^{\text{norm}}, \ldots, S_k^{\text{norm}}) \in \mathbb{R}^{k \times k} \tag{11}$$

The calibrated projection of a centered sample x_c is:

$$z' = AW_k^\top x_c = Az \tag{12}$$

Biometric matching is performed using cosine similarity on the calibrated projections:

$$\text{sim}(z'_p, z'_g) = \frac{(z'_p)^\top z'_g}{\|z'_p\|_2 \|z'_g\|_2} \tag{13}$$

where z'_p is the calibrated probe projection and z'_g is the calibrated gallery template (computed as the element-wise average of the calibrated training projections for each class).

Algorithm 1 XPCA Utility-Guided Calibration

Require: Training data $X \in \mathbb{R}^{N \times D}$, labels y, target dimension k, hyperparameters α, β, γ, bootstrap iterations T
Ensure: Calibrated projections $Z'_{\text{train}}, Z'_{\text{test}}$

```
Fit PCA on X: obtain mean μ, eigenvectors W, eigenvalues λ
Compute variance scores: V_i ← λ_i / Σ_j λ_j for i = 1, ..., k
Project training data: Z_train ← (X − 1μ^⊤) W_k
Compute global Fisher scores D_i on Z_train using Eq. 8
for t = 1 to T do
    Resample indices with replacement → Z^(t), y^(t)
    Compute bootstrapped Fisher scores D_i^(t)
end for
Compute instability: N_i ← std_t(D_i^(t)) / (mean_t(D_i^(t)) + ϵ)
Compute raw utility: S_i ← αV_i + βD_i − γN_i
Min-max normalize: S_i^norm ← (S_i − min_j S_j) / (max_j S_j − min_j S_j + ϵ)
Build calibration matrix: A ← diag(S_1^norm, ..., S_k^norm)
Calibrate: Z'_train ← Z_train A,   Z'_test ← Z_test A
```

E. Algorithmic Pipeline

The complete XPCA pipeline is given in Algorithm 1. Training requires fitting PCA once, computing Fisher scores once, and running T bootstrap iterations. At test time, calibration requires only a single element-wise multiplication, adding negligible computational cost.

F. Computational Complexity Analysis

Training. SVD of $\bar{X} \in \mathbb{R}^{N \times D}$ requires $\mathcal{O}(ND \cdot \min(N, D))$ operations. For the IITD dataset ($N = 1380$, $D = 16{,}384$), this is $\mathcal{O}(N^2 D) \approx 3.1 \times 10^{10}$. Vectorized Fisher score computation on $Z_{\text{train}} \in \mathbb{R}^{N \times k}$ requires $\mathcal{O}(NCk)$ operations, where C is the number of classes. Running $T = 100$ bootstrap iterations requires $\mathcal{O}(TNCk)$ total. Since $T \cdot k \ll D$, the bootstrap overhead is negligible relative to SVD.

Matching. Projecting a raw sample costs $\mathcal{O}(Dk)$. Calibration scaling (element-wise multiplication) costs $\mathcal{O}(k)$. Computing cosine similarity against one gallery template costs $\mathcal{O}(k)$. Searching the entire gallery of C templates costs $\mathcal{O}(Ck)$. The *per-sample* matching complexity is identical for PCA and XPCA.

IV. Experimental Setup

A. Dataset

All experiments are conducted on the **IIT Delhi (IITD) Touchless Palmprint Database (Version 1.0)** [1]. The dataset contains 2,601 grayscale bitmap images from 230 subjects (460 classes, treating left and right hands of each subject as distinct biometric identities). Images are acquired using a touchless setup with controlled indoor illumination. Each image is resized to 128×128 pixels, yielding a raw feature dimensionality of $D = 128 \times 128 = 16{,}384$.

B. Preprocessing

Each flattened image vector $x \in \mathbb{R}^D$ is normalized to $[0, 1]$ by dividing by the maximum pixel intensity (255). Individual sample zero-mean normalization ($x \leftarrow x - \bar{x}$) is then applied to each sample independently to suppress global illumination gradients while preserving local contrast and texture features.

C. Verification Protocol

For each of the $C = 460$ classes, the first 3 sorted images (by filename) are assigned to the training set, and the remaining images form the test set. This yields $N_{\text{train}} = 1{,}380$ training images and $N_{\text{test}} = 1{,}221$ test images. PCA is fitted exclusively on the training set.

For each class, a *gallery template* is computed as the element-wise average of the training projections. Each test probe is matched against all 460 gallery templates using cosine similarity (Eq. 13). The genuine score is the similarity between the probe and its own class template; the 459 impostor scores are the similarities between the probe and all other class templates.

D. Evaluation Metrics

We report three complementary metrics:

- **Equal Error Rate (EER):** The operating point where the False Acceptance Rate (FAR) equals the False Rejection Rate (FRR). Lower EER indicates better verification performance.
- **Balanced Accuracy:** Defined as $\text{Acc} = 1 - \text{EER}$.
- **ROC AUC:** The area under the Receiver Operating Characteristic curve, measuring the overall discriminative power across all thresholds.

E. Experimental Configurations

We evaluate the following configurations:

- **Baseline:** Standard PCA projection with unweighted cosine similarity.
- **XPCA:** Calibrated projection with utility weights $(\alpha, \beta, \gamma) = (0.2, 0.6, 0.2)$.
- **Ablation A (Variance Only):** $S_i = V_i$.
- **Ablation B (Discriminability Only):** $S_i = D_i$.
- **Ablation C (Instability Only):** $S_i = -N_i$.
- **Ablation D (Combined XPCA):** $S_i = 0.2V_i + 0.6D_i - 0.2N_i$.

Ablation experiments are conducted at $k = 128$, the dimension achieving the best baseline EER.

V. Experimental Results

A. PCA vs. XPCA Across Dimensions

Table I presents the verification performance of standard PCA and XPCA with default weights $(\alpha, \beta, \gamma) = (0.2, 0.6, 0.2)$ across five subspace dimensions.

Several patterns are evident:

Consistent degradation. XPCA calibration increases EER at every tested dimension. The absolute EER increase ranges from $+2.84$ percentage points ($k = 512$) to $+5.72$ percentage

TABLE I: PCA vs. XPCA verification performance comparison across subspace dimensions. Bold indicates the best result in each metric column. ΔEER denotes the absolute EER increase from calibration.

Method	k	EER (%)	Acc. (%)	AUC	ΔEER
PCA	32	10.57	89.43	0.9564	—
XPCA	32	16.29	83.71	0.9221	+5.72
PCA	64	9.83	90.17	0.9605	—
XPCA	64	14.51	85.49	0.9357	+4.68
PCA	128	**9.50**	**90.50**	**0.9619**	—
XPCA	128	13.17	86.83	0.9432	+3.67
PCA	256	9.41	90.59	0.9619	—
XPCA	256	12.44	87.56	0.9473	+3.03
PCA	512	9.28	90.72	0.9618	—
XPCA	512	12.12	87.88	0.9483	+2.84

TABLE II: Ablation study at $k = 128$. PCA baseline EER is 9.50%. All calibrated configurations degrade performance.

Configuration	EER (%)	Acc. (%)	AUC
PCA Baseline (no calibration)	**9.50**	**90.50**	**0.9619**
A: Variance Only ($S_i = V_i$)	23.27	76.73	0.8610
B: Discriminability Only ($S_i = D_i$)	13.18	86.82	0.9437
C: Instability Only ($S_i = -N_i$)	11.06	88.94	0.9558
D: Combined XPCA	13.17	86.83	0.9432

points ($k = 32$). This rules out the possibility that calibration helps at certain dimensionalities.

Degradation decreases with k. The EER gap narrows as k increases: ΔEER drops from 5.72 at $k = 32$ to 2.84 at $k = 512$. This is consistent with the coordinate collapse mechanism (Section VI): at higher k, more components are included, partially mitigating the dominance of a few high-utility coordinates.

Baseline saturation. Standard PCA EER saturates near $k = 128$ (9.50%), with marginal improvement at $k = 256$ (9.41%) and $k = 512$ (9.28%). This saturation indicates that the useful discriminative information in the PCA subspace is largely captured by the first 128 components.

B. Ablation Study ($k = 128$)

To isolate the contribution of each utility component and to determine whether the failure stems from a particular signal, we perform an ablation study at $k = 128$. Table II presents the results.

All calibration configurations degrade performance. No combination of utility signals improves over the unweighted PCA baseline. This confirms that the failure is structural—it is the act of non-uniform diagonal scaling itself, not the choice of which signal to use, that degrades performance.

Variance-only weighting suffers the worst collapse. Configuration A (EER = 23.27%) nearly triples the baseline EER. This is expected from the coordinate collapse theory: eigenvalue variance scores decay exponentially ($V_1 = 0.1178$, $V_{128} = 0.00096$), producing an extreme range in the scaling

TABLE III: Hyperparameter sweep at $k = 128$. Varying weight configurations has negligible effect on EER.

α	β	γ	EER (%)	Acc. (%)	AUC
0.2	0.6	0.2	13.17	86.83	0.9432
0.3	0.5	0.2	13.15	86.85	0.9427
0.4	0.4	0.2	13.35	86.65	0.9420

coefficients and hence an extremely ill-conditioned metric tensor.

Instability-only weighting is least harmful. Configuration C (EER = 11.06%) incurs the smallest degradation. Bootstrap instability scores N_i are relatively uniform across components (coefficient of variation ≈ 0.08–0.13), keeping the condition number $\kappa(M)$ small relative to configurations A and B.

Fisher-only and Combined produce similar degradation. Configurations B (13.18%) and D (13.17%) are nearly identical, indicating that the combined score is dominated by the Fisher term (which has the largest weight $\beta = 0.6$).

C. Hyperparameter Sweep ($k = 128$)

To verify that the failure is not an artifact of a particular weight configuration, we test three different (α, β, γ) settings. Table III shows the results.

The EER varies by only ± 0.10 percentage points across weight configurations (range: 13.15%–13.35%). This minimal sensitivity confirms that the degradation is not a hyperparameter tuning issue but a *structural limitation* of the diagonal calibration framework under cosine matching.

D. Inference Latency

We measure the average CPU time to project a single sample image ($D = 16{,}384 \to k = 128$) over 1,000 repetitions:

- **Standard PCA:** 1.364 ms per sample.
- **XPCA (calibrated):** 0.648 ms per sample.

The XPCA calibrated projection is faster because it uses direct NumPy matrix multiplication rather than scikit-learn's `transform()` wrapper, which incurs object overhead. The element-wise calibration scaling adds $\mathcal{O}(k)$ operations, which is negligible relative to the $\mathcal{O}(Dk)$ projection cost.

VI. Failure Analysis: Coordinate Collapse

The experimental results demonstrate that utility-guided diagonal calibration degrades verification performance across all tested configurations. In this section, we derive the mathematical mechanism responsible for this failure.

A. Quadratic Amplification Under Cosine Similarity

Let $z, u \in \mathbb{R}^k$ be the standard PCA projections of two samples. Under XPCA, the calibrated projections are $z' = Az$ and $u' = Au$, where $A = \mathrm{diag}(a_1, \ldots, a_k)$ with $a_i = S_i^{\mathrm{norm}}$. Substituting into the cosine similarity formula:

$$\mathrm{sim}(z', u') = \frac{(Az)^\top (Au)}{\|Az\|_2 \|Au\|_2} = \frac{z^\top A^\top A\, u}{\sqrt{z^\top A^\top A\, z} \cdot \sqrt{u^\top A^\top A\, u}} \tag{14}$$

Since A is diagonal, $A^\top A = A^2 = \text{diag}(a_1^2, \ldots, a_k^2)$. Expanding:

$$\text{sim}(z', u') = \frac{\sum_{i=1}^{k} a_i^2 \, z_i u_i}{\sqrt{\sum_{i=1}^{k} a_i^2 \, z_i^2} \cdot \sqrt{\sum_{i=1}^{k} a_i^2 \, u_i^2}} \tag{15}$$

Proposition 1 (Quadratic Weight Amplification). *Diagonal scaling by $A = diag(a_1, \ldots, a_k)$ under cosine similarity is equivalent to computing unscaled cosine similarity with respect to the weighted inner product defined by the metric tensor $M = A^2 = diag(a_1^2, \ldots, a_k^2)$. The effective weight of coordinate i is a_i^2, not a_i.*

This quadratic amplification is the root cause of coordinate collapse. If the utility scores a_i span a dynamic range of $r = a_{\max}/a_{\min}$, the effective weights span a range of r^2. For the XPCA utility scores on the IITD dataset, the component with the highest normalized score has $S_{\max}^{\text{norm}} = 1.0$ while the component with the lowest has $S_{\min}^{\text{norm}} \approx \epsilon \approx 10^{-10}$ (due to min-max normalization). The effective dynamic range is therefore $r^2 \approx 10^{20}$.

B. Metric Tensor Ill-Conditioning

Definition 1 (Induced Metric Tensor). *The cosine similarity under diagonal calibration A is equivalent to unweighted cosine similarity under the coordinate transformation induced by the Mahalanobis-like metric tensor:*

$$M = A^\top A = diag(a_1^2, a_2^2, \ldots, a_k^2) \tag{16}$$

Theorem 1 (Condition Number Divergence). *Let the utility scores be min-max normalized to $[0, 1]$ via Eq. 10. Then:*

$$\kappa(M) = \frac{\max_i a_i^2}{\min_i a_i^2} = \frac{(S_{\max}^{norm})^2}{(S_{\min}^{norm})^2} = \frac{1}{\epsilon^2} \to \infty \tag{17}$$

where ϵ is the numerical regularization constant in the min-max normalization denominator.

Proof. By construction of min-max normalization (Eq. 10), $S_{\max}^{\text{norm}} = 1$ and $S_{\min}^{\text{norm}} = \epsilon/(\text{range} + \epsilon) \leq \epsilon$ for any nonzero range. The eigenvalues of the diagonal metric tensor M are $\{a_i^2\}$. The condition number is the ratio of the largest to the smallest eigenvalue:

$$\kappa(M) = \frac{1^2}{\epsilon^2} = \frac{1}{\epsilon^2} \tag{18}$$

For $\epsilon = 10^{-10}$, $\kappa(M) = 10^{20}$, indicating extreme ill-conditioning. □

C. Effective Dimensionality Collapse

Corollary 1 (Coordinate Collapse). *As $\kappa(M) \to \infty$, the cosine similarity degenerates to depend only on the coordinates with the largest a_i^2 values. In the limit:*

$$sim(z', u') \to \frac{a_{(1)}^2 z_{(1)} u_{(1)}}{|a_{(1)}^2 z_{(1)}^2|^{1/2} |a_{(1)}^2 u_{(1)}^2|^{1/2}} = sign(z_{(1)} u_{(1)}) \tag{19}$$

where subscript (1) denotes the coordinate with the largest utility weight. The effective dimensionality of the embedding collapses from k to 1.

This collapse is catastrophic for biometric verification. The class-discriminative information is distributed across multiple principal components—our component ablation analysis shows that removing any single top-10 component increases EER by at most 0.83%, confirming that no single component carries sufficient discriminative power alone. Coordinate collapse destroys this distributed information by forcing the similarity computation to be determined by a single coordinate.

D. Experimental Validation of Collapse Theory

The ablation results in Table II provide direct empirical validation of the coordinate collapse theory:

Variance-only (A) exhibits the worst collapse. The normalized eigenvalue variance scores V_i decay exponentially: $V_1 = 0.1178$, $V_{10} = 0.0148$, $V_{50} = 0.0030$, $V_{128} = 0.0010$. After min-max normalization, the dynamic range of the scores is approximately $118 : 1$, yielding a squared dynamic range of approximately $13{,}924 : 1$. This extreme ill-conditioning produces EER = 23.27%.

Fisher-only (B) produces moderate collapse. Fisher scores are non-uniform but less extreme than variance scores. The top Fisher component (PC 1, $D_1 = 4.64$) is approximately $20\times$ larger than the median Fisher score. After squaring, the dynamic range is $\sim 400 : 1$, producing EER = 13.18%.

Instability-only (C) produces the mildest collapse. Bootstrap instability coefficients N_i cluster tightly around 0.08–0.13 across most components. The dynamic range is approximately $1.6 : 1$, yielding a squared range of $\sim 2.6 : 1$. This mild non-uniformity produces only moderate degradation (EER = 11.06%), consistent with a nearly isotropic metric tensor.

Remark 1. *The ordering of ablation EERs ($A \gg B \approx D > C >$ baseline) perfectly matches the predicted ordering based on the condition number $\kappa(M)$ of each configuration's metric tensor: $\kappa_A \gg \kappa_B \approx \kappa_D > \kappa_C > \kappa_{baseline} = 1$.*

E. Why the Degradation Decreases with k

The observation that ΔEER decreases as k increases (Table I) is also explained by the collapse mechanism. At larger k, the Fisher scores of the additional lower-variance components are more uniformly distributed, partially reducing the condition number of the metric tensor. Furthermore, at larger k, the contribution of the dominant components to the total summation is diluted by the sheer number of additional coordinates, weakening the dominance effect.

VII. Discussion

A. Feature Importance vs. Representation Geometry

The coordinate collapse finding reveals a fundamental distinction between two concepts that are often conflated in the feature engineering literature:

Feature importance refers to which components carry discriminative information for a classification or verification task. Feature importance is a *task-level* property that can be measured via Fisher ratios, mutual information, or ablation studies.

Representation geometry refers to how coordinates must be scaled to preserve the distance or similarity relationships that the matching metric requires. Representation geometry is a *metric-level* property that depends on the mathematical structure of the matching function.

The XPCA framework conflates these two concepts by using feature importance scores (Fisher discriminability) directly as geometric scaling factors. This conflation fails because cosine similarity requires *isotropic* coordinate contributions—all coordinates must contribute equally to the angle computation for the full angular information to be preserved. Applying non-uniform scaling distorts the angles, effectively "rotating" the template toward the high-utility axes and destroying the angular relationships that encode biometric identity.

B. Connection to Metric Learning

The diagonal calibration $z' = Az$ followed by cosine similarity is mathematically equivalent to computing a Mahalanobis-like cosine similarity under the metric tensor $M = A^2$. This establishes a direct connection between XPCA and the family of diagonal metric learning methods. The coordinate collapse result implies that any diagonal metric learning method that uses cosine similarity as the base metric must ensure that the learned diagonal matrix M has a bounded condition number. Specifically, the condition number constraint:

$$\kappa(M) = \frac{M_{\max}}{M_{\min}} \leq \kappa_{\max} \tag{20}$$

should be added as an explicit regularization constraint in the metric learning objective.

C. Implications for Biometric System Design

The coordinate collapse result provides a concrete design principle for biometric systems:

Principle. When using cosine similarity (or any angular metric) for matching, coordinate scaling factors must satisfy:

$$\frac{\max_i a_i}{\min_i a_i} \leq R_{\max} \tag{21}$$

where $R_{\max}$ is a finite bound. As a rule of thumb, $R_{\max} \leq 10$ ensures that the squared dynamic range $R_{\max}^2 \leq 100$, keeping the metric tensor well-conditioned.

D. Scientific Value of Negative Results

This work contributes to the growing body of negative results in machine learning [10]. The coordinate collapse mechanism is not specific to palmprint verification or to PCA—it applies to *any* system that combines non-uniform coordinate scaling with angular similarity matching. This generality makes the finding relevant to a broad audience, including practitioners working on face, iris, or gait recognition systems that use subspace projections with cosine matching.

VIII. Future Work

The coordinate collapse mechanism provides clear guidance for designing calibration schemes that avoid ill-conditioning. We outline four regularized alternatives.

A. Softmax Scaling

Replace min-max normalization with temperature-controlled softmax:

$$a_i = \frac{\exp(S_i/\tau)}{\sum_j \exp(S_j/\tau)} \tag{22}$$

Since $\exp(\cdot) > 0$ for all finite arguments, all weights are strictly positive, and the condition number is bounded:

$$\kappa(M) = \exp\left(\frac{2(S_{\max} - S_{\min})}{\tau}\right) \tag{23}$$

By choosing τ sufficiently large, the condition number can be controlled to any desired level.

B. Residual Calibration

Define the scaling as a small perturbation around unity:

$$a_i = 1 + \eta \cdot S_i^{\text{norm}} \tag{24}$$

where $\eta \geq 0$ is a small scaling parameter. This ensures that $a_i \in [1, 1+\eta]$, giving:

$$\kappa(M) = \left(\frac{1+\eta}{1}\right)^2 = (1+\eta)^2 \tag{25}$$

For $\eta = 0.1$, $\kappa(M) = 1.21$, which is well-conditioned. The parameter η interpolates between unweighted PCA ($\eta = 0$) and full calibration ($\eta \to \infty$).

C. Rank-Based Calibration

Replace raw utility scores with a function of the utility *rank*:

$$a_i = f\left(\text{rank}(S_i)\right) \tag{26}$$

where f is a monotonically decreasing function such as $f(r) = k^{-1/p} \cdot r^{-1/p}$ for some $p > 0$. Rank-based weights eliminate the exponential decay of eigenvalue-based scores and produce a controlled dynamic range.

D. Condition-Number-Constrained Metric Learning

Learn the diagonal metric tensor $M = \text{diag}(m_1, \ldots, m_k)$ by optimizing a verification loss subject to explicit spectral constraints:

$$\min_M \mathcal{L}_{\text{verif}}(M) \quad \text{s.t.} \quad \frac{\max_i m_i}{\min_i m_i} \leq \kappa_{\max}, \quad m_i > 0\ \forall i \tag{27}$$

This formulation preserves the computational efficiency of diagonal scaling while provably preventing coordinate collapse.

IX. Conclusion

This paper investigated XPCA, a utility-guided diagonal calibration framework for PCA-based biometric verification. The framework computes additive utility scores integrating explained variance, Fisher class-discriminability, and bootstrap stability for each principal component, and applies the resulting weights as a diagonal scaling matrix prior to cosine similarity matching.

Evaluating the framework on the IIT Delhi touchless palmprint database across five subspace dimensions, we documented the consistent negative result that utility-based calibration degrades verification performance at every tested configuration. At the optimal baseline dimension $k = 128$, calibration increases the Equal Error Rate from 9.50% to 13.17%. A four-configuration ablation study and three-configuration hyperparameter sweep confirmed that this degradation is structural rather than attributable to suboptimal parameter choices.

We derived the mathematical mechanism responsible for this failure: *coordinate collapse*. Under cosine similarity, diagonal scaling factors appear quadratically (a_i^2), amplifying any non-uniformity in the utility scores. When combined with min-max normalization (which forces at least one weight toward zero), the induced metric tensor becomes ill-conditioned ($\kappa(M) \to \infty$), effectively collapsing the k-dimensional embedding to a low-dimensional manifold dominated by a few high-utility coordinates. This destroys the distributed biometric information encoded across the full principal subspace.

The coordinate collapse mechanism is not specific to palmprint verification or to PCA—it applies to any system that combines non-uniform coordinate scaling with angular similarity matching. This generality makes the finding a transferable design principle: diagonal calibration under cosine similarity requires explicit condition-number regularization to prevent representational collapse.

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